

AlphaZero for Protein Engineering

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Protein engineering confronts the challenge of navigating vast sequence spaces to discover functional proteins. While reinforcement learning (RL) offers a promising framework where an agent optimizes sequences through mutations against structural/functional rewards, existing approaches like EvoPlay suffer from limited feature extraction due to their reliance on 1D convolutional neural networks (CNNs), which fail to capture long-range residue dependencies or leverage semantic representations. To overcome this, we enhance EvoPlay by integrating ESM2, a Transformer-based protein language model pretrained via masked language modeling. Our framework replaces 1D CNNs with ESM2’s self-attention mechanism to extract semantically rich, structure-aware embeddings; redesigns the agent and environment using ESM2’s multimodal features (mapped via linear layers to action-space probabilities and rewards); and demonstrates superior *in silico* performance, generating sequences with higher functional scores than the original EvoPlay. This work advances computational protein design by unifying ESM2’s representation power with RL-driven optimization.

1 INTRODUCTION

Proteins are the main carriers of life activities, and their biological functions are mainly determined by their structures, which in turn are determined by their amino acid sequences. With the development of protein structure prediction and determination techniques, researchers have begun to focus on designing new protein sequences to obtain proteins with specific functions, a field known as protein engineering. Proteins typically contain hundreds to thousands of amino acid residues with 20 possible choices at each position, resulting in a sequence design space of well over 20^{1000} , much higher than the total number of atoms in the observable universe (10^{80}). How to find proteins with specific functions in such a large space is the central problem facing the field.

Reinforcement learning, a powerful AI approach that maximizes cumulative rewards by guiding intelligences to learn by trial and error with their environment, provides a powerful tool for solving such combinatorial optimization problems. As shown in Fig. 1, in the framework of applying reinforcement learning to the problem of protein sequence design:

- **Agent** is an algorithmic model (e.g., neural network) that learns a sequence optimization strategy.
- **State** is the specific protein sequence currently being evaluated or modified.
- **Action** is an action that specifies a change in the type of amino acid at a particular position in the sequence (e.g., replacing the amino acid at position 50 with an alanine). In the field of protein engineering, it is called mutation.
- **Environment** is a computational module that receives new sequences (action results) generated by an agent, predicts its 3D structure using a structure prediction tool, evaluates the structure’s properties (e.g., stability, functional activity) by applying a predefined scoring function, and ultimately feeds the computed scoring value to the agent as a Reward.
- **Reward** is the output value of the scoring function, reflecting the degree to which the current sequence satisfies the design goals.

The goal of an agent is to learn a strategy to discover high-performance protein sequences by interacting with the environment (generating and evaluating sequences) to maximize cumulative rewards.

EvoPlay [15] is the pioneering work for exploring model-based reinforcement learning in protein engineering. It is a single-player optimization framework inspired by AlphaZero [10] self-play reinforcement learning, built for goal-directed protein design. At its core, it performs directed evolution (DE) in sequence space by simulating a *in silico* mutating agent. The algorithm is benchmarked in a simulation environment, using AlphaFold2 [2] as a structural surrogate to optimize sequences by structural reward at the protein-peptide interface.

In order to build both the agent (for sequence optimization) and the environment (for reward generation based on sequence), EvoPlay first performed feature extraction on the input protein sequences. However, due to technological constraints at the time, this approach has the following limitations: the authors did not adopt the self-attention mechanism from Transformers to capture long-range dependencies between amino acid residues, nor did they leverage any pre-trained model to derive semantic representations of the sequences. Instead, they relied solely on a basic one-dimensional convolutional neural network (CNN) for feature extraction. In contrast, a Transformer with self-attention can model interactions among amino acids across the entire sequence, whereas a standard 1D convolution only captures local information and struggles to learn associations between distant residues. Furthermore, compared to sequence models that are pre-trained via a Masked Language Modeling objective and then fine-tuned, a 1D convolution network trained from scratch is limited in its ability to obtain richly semantic sequence representations and cannot fully uncover the latent structural and functional information embedded in protein sequences.

In this study, we employ the Transformer-based sequence model ESM2 [3], which is pretrained via a Masked Language Model, to extract features from input protein sequences. By leveraging the self-attention mechanism and Masked Language Model pretraining, ESM2 can derive embeddings rich in biological semantics from the sequences. Moreover, ESM2 serves as the foundation for ESMFold [3], a protein structure prediction model, indicating that its representations capture not only the semantics of the one-dimensional sequence but also features of the three-dimensional structure. Through its Masked Language Model pretraining, ESM2 effectively constructs a shared embedding space for protein sequence and structure[4]. We feed the multimodal features extracted by ESM2 into a shallow linear fully connected layer, mapping them respectively to reward values or action-space probabilities, thereby forming our agent and environment models. By utilizing the deep features from ESM2, we are able to generate protein sequences that outperform the original EvoPlay in functional metrics; these embeddings also enhance the agent’s learning of sequence-optimization strategies and the construction of the environment model.

Our contributions are summarized as follows:

- We introduce a Transformer-based sequence modeling pipeline by integrating the pre-trained ESM2 model into both the agent and environment of EvoPlay. Unlike the original approach, which relied on a 1D convolutional network, our ESM2-driven feature extractor leverages self-attention and masked language modeling to capture both semantic and structural information from protein sequences.
- We construct a novel environment model that maps ESM2’s multimodal embeddings to reward values, and we design an updated agent network that projects ESM2 features into action-space probabilities and state values. This design enables more effective exploration of sequence space and more accurate feedback, ultimately guiding the agent toward proteins with high-functionality.

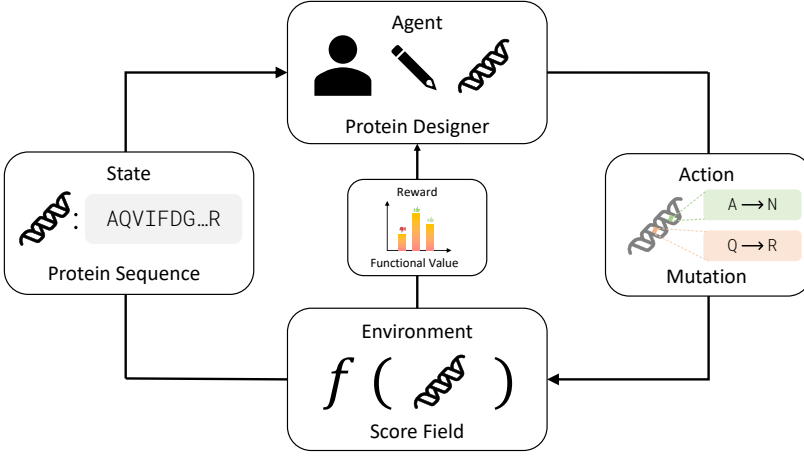


Fig. 1. Reinforcement learning framework for protein sequence design.

- We demonstrate *in silico* that sequences generated using our ESM2-enhanced framework achieve higher functional scores compared to the original EvoPlay method.

2 RELATED WORKS

2.1 AlphaZero and Its Variants

AlphaZero [10] is a landmark reinforcement learning (RL) algorithm developed by DeepMind that demonstrates the power of self-play and Monte Carlo Tree Search (MCTS) in complex decision-making domains. As introduced by Silver et al. [10], AlphaZero learned to master the games of chess, shogi, and Go solely through reinforcement learning, without any human data, relying only on the rules of the games. The core of AlphaZero combines a deep neural network that outputs both a value estimate and a policy distribution with a powerful search procedure: MCTS guided by the predictions of the neural network.

Several works have extended or adapted the AlphaZero paradigm beyond board games. For example, MuZero [9] eliminated the need to know the dynamics of the environment, instead learning a latent model of the environment that supports planning. In protein engineering, the rules of environmental dynamics are known and determined due to the validity of amino acid mutations (arbitrary mutations are legal). At this point, MuZero’s underlying state learning mechanism can optimize planning efficiency by constructing compact sequence representations and fast reward prediction models that do not simply degenerate into AlphaZero.

The strengths of AlphaZero and its variants, namely, its model-based planning, deep neural guidance, and self-improvement through simulation, make it an attractive candidate for challenging structured domains such as protein design. In this work, we build on the AlphaZero/MuZero foundation, adapting its architecture and training loop to operate in a biological sequence space rather than a traditional game setting.

2.2 Protein Representation

In recent years, protein computational models have made remarkable breakthroughs in the field of structure prediction and design, which are mainly categorized into two types of paradigms. The

first one is structural prediction and characterization model, which includes ESM2 [3], a large-scale protein language model, FoldSeek [14], a 3D structural tokenization method, and SaProt [11], a structure-aware vocabulary method that integrates the above techniques, and the second one is generative design model, represented by ESM3, which is based on the joint modeling of sequence-structure-function relationship by VQ-VAE [13] codebook. ESM2/FoldSeek/SaProt provides fast structure evaluation capability for reinforcement learning agents; while the structure encoding of ESM3 [1] (replacing amino acid sequences with discrete symbols) forms a new type of state representation whose mechanism matches the implicit state design concept of MuZero.

2.3 RL in Protein Design

Recent advances [5, 8, 15] in deep learning and protein structure prediction have driven growing interest in applying RL to protein design. Given the vast and complex search space of amino acid sequences, RL offers a promising alternative to traditional methods by enabling efficient, policy-driven exploration.

One representative approach is EvoPlay [15], a self-play reinforcement learning framework inspired by AlphaZero. EvoPlay treats protein sequence optimization as a single-player game, where residue mutations are modeled as actions. A policy-value neural network interacts with MCTS, enabling the agent to balance breadth and depth in sequence exploration. EvoPlay supports both functional and structural evaluation, including integration with AlphaFold2 as a surrogate model. It achieves superior performance in multiple *in silico* protein engineering tasks and has been experimentally validated by designing luciferase variants with up to a 7.8-fold increase in bioluminescence.

Beyond EvoPlay, Lutz et al. [5] proposed a top-down RL approach that uses MCTS to sample protein conformations under global architectural and functional constraints, effectively tackling complex structural design tasks. Renard et al. [8] extended MCTS-based backbone design by introducing threshold-based rewards and secondary objectives, significantly improving design precision. Subramanian et al. [12] explores combining reinforcement learning with protein language models (PLMs) for sequence design. Instead of relying on structural scores, PLMs are used as reward functions to guide mutation policies toward biologically plausible sequences. To reduce the computational cost of large PLMs, a smaller proxy model is co-trained with the RL agent. Experiments show that this approach generates diverse and biologically meaningful sequences, highlighting the potential of RL+PLM frameworks in protein design.

3 METHODS

In this section, we describe the key components of our approach to protein sequence optimization. In Sec. 3.1, we review the EvoPlay framework and its MCTS workflow, detailing how each state and action is defined and how surrogate rewards guide exploration. Then, in Sec. 3.2, we introduce our use of the pretrained ESM2 Transformer to obtain both residue and sequence level embeddings, highlighting how these rich representations replace one-hot encodings. Finally, in Sec. 3.3, we explain how we integrate ESM2 embeddings into the surrogate and policy-value networks, and we outline the modified network architectures. The overview of our adaptation to EvoPlay is shown in Fig. 2.

3.1 Overview of EvoPlay and MCTS workflow

In our adaptation of EvoPlay for protein sequence design, we follow EvoPlay’s definition. Each state s_t is encoded as a binary matrix of size $20 \times L$, where L is the sequence length and each of the 20 rows corresponds to one of the canonical amino acids in one-hot form. The action space is the flattened equivalent of this matrix: a vector of length $20L$. Selecting an action a thus corresponds

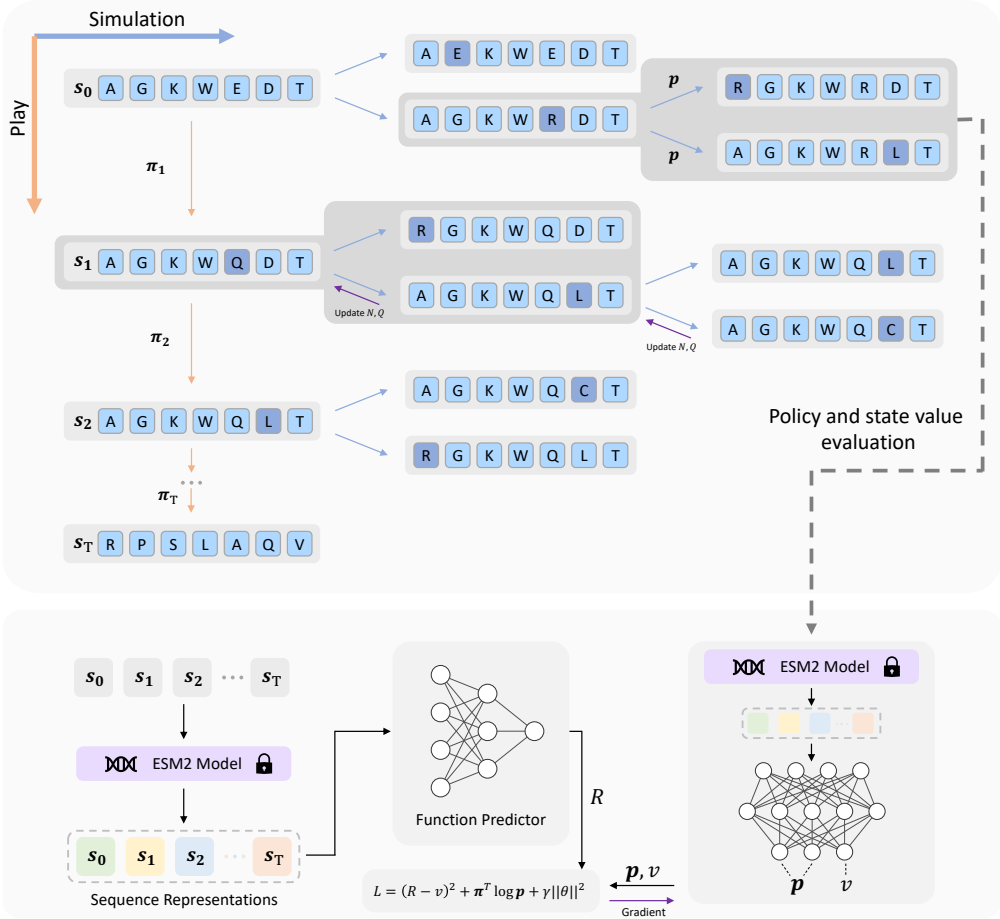


Fig. 2. Overview of our modified EvoPlay framework incorporating ESM2 embeddings. Instead of using one-hot encoded sequences, we adopt pretrained sequence embeddings from ESM2. The surrogate model takes the sequence-level embedding to predict functional rewards, while the policy-value network uses residue-level embeddings to produce action priors and value estimates.

to introducing a single-site mutation at a particular position; specifically, the chosen entry in the state matrix is set to 1, and the remaining entries in its column are reset to 0. The invalid moves, including introducing no mutation on the sequence or returning to previously explored sequences, are removed during each episode. An episode consists of a sequence of moves $\{a_1, a_2, \dots, a_T\}$ producing intermediate states $\{s_1, s_2, \dots, s_T\}$ and receiving surrogate rewards $\{r_1, r_2, \dots, r_T\}$. Episodes terminate either when the surrogate score decreases relative to the previous step ($r_t < r_{t-1}$).

In our workflow, the surrogate model g_ϕ serves as a fast, learned approximation of the true fitness landscape, allowing us to evaluate candidate sequences without costly wet-lab or high-resolution structural calculations. Concretely, we passed sequence state through a small surrogate

model parameterized by ϕ , which outputs a scalar surrogate reward

$$r_t = g_\phi(s_t). \quad (1)$$

The parameters ϕ of the surrogate are learned by minimizing a mean squared error loss between the model’s predictions and ground-truth fitness scores r_i^{true} from wet-lab:

$$\mathcal{L}(\phi) = \frac{1}{N} \sum_{i=1}^N (g_\phi(s_i) - r_i^{\text{true}})^2. \quad (2)$$

The surrogate model g_ϕ provides the surrogate rewards $\{r_1, r_2, \dots, r_T\}$, guiding the search.

During search, MCTS is guided by a policy-value network f_θ . Each simulation starts at the root node s_{root} and descends the tree by selecting at each node s the child action a that maximizes the sum of its current action value $Q_{s,a}$ and an exploration bonus $U_{s,a}$:

$$a^* = \arg \max_a (Q_{s,a} + U_{s,a}), \quad (3)$$

where

$$U_{s,a} = c \cdot P_{s,a} \frac{\sqrt{N_{s,a}^{\text{Parent}}}}{1 + N_{s,a}}. \quad (4)$$

Here, $P_{s,a}$ is the prior probability of selecting a in state s , $N_{s,a}$ is the visit count of edge (s, a) , and c balances exploration and exploitation. When a leaf node s_L is reached, the tree is expanded by evaluating $f_\theta(s_L)$ to obtain both a value estimate v and a vector of action priors $\{P_{s_L,a}\}$, which initialize its outgoing edges. Backpropagation then updates Q and N along the path from s_L back to s_{root} :

$$N_{s,a} \leftarrow N_{s,a} + 1, \quad Q_{s,a} \leftarrow Q_{s,a} + \frac{v - Q_{s,a}}{N_{s,a}}. \quad (5)$$

After completing a predefined number of simulations at time step t , a final move π_t is sampled from the root’s visit counts according to

$$\pi_t(a) \propto N(s_t, a)^{1/\tau}, \quad (6)$$

where τ is a temperature parameter that controls exploration. The resulting triplet (s_t, π_t, r_t) is stored for training the network. The policy-value network $f_\theta(s_t)$ outputs both a probability distribution over actions $P_{s_t,a}$ and a scalar value v_{s_t} , and is trained to minimize the combined loss

$$\mathcal{L}(\theta) = \sum_t \left[(r_t - v_{s_t})^2 - \pi_t^\top \log P_{s_t,a} \right] + \gamma \|\theta\|^2, \quad (7)$$

where γ is an regularization coefficient.

3.2 Sequence Representation with ESM2

To leverage rich, pretrained protein sequence features, we employ the ESM2 model [3] as our embedding backbone. Given an input amino acid sequence of length L , ESM2 first tokenizes each residue into a fixed vocabulary of size 20, representing the 20 canonical residues. The token sequence

$$\{x_1, x_2, \dots, x_L\}, \quad x_i \in \{0, \dots, 19\} \quad (8)$$

is passed through a stack of Transformer encoder layers. At each layer ℓ , the hidden representation for position i is denoted

$$h_i^{(\ell)} \in \mathbb{R}^d, \quad (9)$$

where d is the model’s hidden dimension.

For the residue-level embeddings, we extract the final-layer outputs

$$H = [h_1^{(\ell_{\text{final}})}, h_2^{(\ell_{\text{final}})}, \dots, h_L^{(\ell_{\text{final}})}] \in \mathbb{R}^{L \times d}. \quad (10)$$

These residue-level embeddings capture context-aware biochemical features and are used to inform local mutation decisions.

For global embedding, which is at the sequence level, we apply mean pooling across all positions:

$$\bar{h} = \frac{1}{L} \sum_{i=1}^L h_i^{(\ell_{\text{final}})} \in \mathbb{R}^d. \quad (11)$$

These sequence-level embeddings capture the global biochemical features, which can be used to inform inter-sequence information.

3.3 Integration of ESM2 Sequence Representation with MCTS

In the original EvoPlay workflow, there are two neural-network components: One is called the surrogate model or the environmental model, which takes a one-hot encoding of the protein sequence and predicts a surrogate reward at each step of the episode. The other is the policy-value network, which takes the same one-hot sequence encoding and outputs both a prior distribution over the action space and a value estimate for the current state.

We replace the one-hot encoding inputs to both models with embeddings from ESM2, leveraging its pretrained sequence representations.

For the surrogate model integration, because the surrogate model captures global, functional information about the protein sequence, we feed it the sequence-level embedding $\bar{h}$ rather than a one-hot matrix. After extraction, $\bar{h}$ is passed through a shallow linear module to produce the surrogate reward prediction.

The policy-value network requires local, position-specific information to model the effects of single-site mutations and to generate action priors. Here we use the residue-level embeddings H . The residue-level embedding H is also mapped via a shallow linear module to match the original output dimensionality, which produces the prior distribution $P_{s,a}$ and value estimate v_s .

In the original EvoPlay, both networks applied a stack of 1D convolutional layers directly to the one-hot state matrix and then a linear output module. In our implementation, after obtaining ESM2 embeddings ($\bar{h}$ or H), we replace the convolutional stack and the linear modules directly with linear heads. This design is simple yet effective, fully leveraging ESM2’s rich sequence representations.

The training objectives for both the surrogate model and the policy-value network are identical to those used in the original EvoPlay framework. In that work, the surrogate model’s training set is progressively expanded to mitigate overfitting when employing a CNN-based architecture. In contrast, by leveraging ESM2-derived protein sequence embeddings, we inherently improve our model’s generalization and reduce its tendency to overfit. Consequently, we train our surrogate model on the entire dataset from the outset.

4 RESULTS

4.1 Dataset and Evaluation Metrics

We use only the PAB1 dataset (UniProt ID P04147), which comprises 36 523 single- and multi-point mutants of length 75. Each mutant is annotated with its XY Enrichment score as the ground-truth fitness label. Similar to the original EvoPlay setup, we do not split PAB1 into training/validation/test. Our surrogate model is trained on the entire dataset to fully leverage all available sequence–fitness pairs.

As our *in silico* oracle, we employ the TAPE model [6] (fine-tuned on PEX data [7]) to predict fitness for any candidate sequence. For each design method, we query the oracle to get the fitness of every proposed mutant. All methods share the same total number of query times to ensure a fair, equal-cost comparison.

We assess structural similarity between our designed variants and those characterized in the wet lab using the orthogonal prediction model AlphaFold2. To ensure that conformational differences arise solely from each input sequence—and not from external alignment or template information—we disable both multiple sequence alignment searches and template searches during the structural prediction.

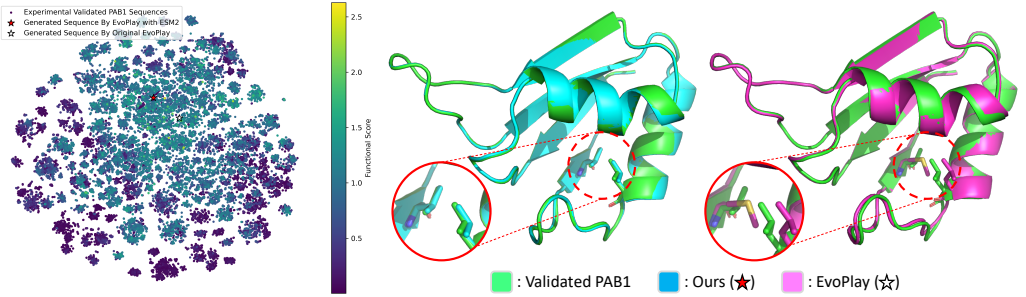


Fig. 3. Visualization of ESM2 embedding space with functional score and structural alignment. Left: ESM2 embeddings of tested sequences show high-fitness clustering, with our method and EvoPlay both generating top sequences in this region. Right: Predicted structures of top sequences show that our method aligns more closely to the best experimental structure in key regions, highlighting improved structural fidelity.

Table 1. Predicted fitness values for 20 sequences generated by the ESM2-augmented method versus those produced by the original EvoPlay. The mean fitness score for sequences from our method is 0.836, compared with 0.771 for EvoPlay’s sequences. The optimal fitness value is highlighted in boldface.

Our method	EvoPlay
1.178	1.144
1.094	1.072
0.875	0.903
0.790	0.861
0.774	0.702
0.762	0.698
0.735	0.677
0.721	0.657
0.716	0.509
0.713	0.484

4.2 Performance on PAB1 protein family

We compared both the quality and performance metrics of our ESM2-enhanced surrogate model and policy-value network against the original EvoPlay.

As shown in Fig. 3, the left panel visualizes ESM2 embeddings (dimensionality-reduced) of experimentally tested sequences, with point colors indicating their measured fitness. High-fitness sequences cluster toward the upper-center region, demonstrating that the ESM2 representations inherently capture information relevant to PAB1 fitness. We overlaid star markers to show where the top sequences generated by our method and by standard EvoPlay fall, and both sequences lie within the high-fitness cluster. In the right panel, we align and visualize the predicted structures of the highest-fitness experimental sequence alongside the top sequences from our method and from EvoPlay. Overall, both two align closely to the experimentally determined structure, but in certain regions (highlighted by red dashed circles), our sequence adhere more precisely to the best wet lab sequence than those produced by EvoPlay. These results indicate that integrating ESM2 embeddings substantially enhances exploration of the protein sequence space.

In terms of metrics, Tab. 1 lists the ten sequences generated by our method alongside ten from the original EvoPlay, and we’ve listed their predicted fitness scores from an *in silico* oracle model. By comparison, our ESM2-augmented approach both explores more broadly and yields sequences with substantially higher predicted fitness. Moreover, the average fitness of our ten sequences exceeds that of EvoPlay’s, demonstrating that incorporating ESM2 embeddings significantly enhances MCTS’s ability to navigate the protein sequence space.

5 DISCUSSION

Protein design involves searching an immense combinatorial space, where efficient exploration and exploitation of sequence-function relationships are crucial. While traditional sampling methods like Markov Chain Monte Carlo (MCMC) have been applied to protein design, their reliance on local stochastic moves limits their effectiveness in this high-dimensional, rugged fitness landscape. MCMC lacks the ability to strategically plan or incorporate learned global sequence representations, which can cause slow convergence and difficulty escaping local optima.

AlphaZero-style reinforcement learning frameworks overcome these challenges by combining policy learning with lookahead search, enabling more directed exploration and decision-making. However, existing implementations such as EvoPlay employ relatively simple 1D convolutional neural networks for sequence feature extraction, which inadequately capture long-range residue interactions and semantic information critical for protein function.

Our work enhances this paradigm by integrating ESM2, a Transformer-based pretrained protein language model, replacing 1D CNNs with its powerful self-attention mechanism. This integration enables extraction of rich, semantically meaningful embeddings that reflect structural and functional constraints. By redesigning the agent’s policy and reward models to leverage these multimodal features, our framework demonstrates improved optimization performance *in silico*, generating sequences with higher functional scores compared to the original EvoPlay.

In summary, although MCMC offers a straightforward approach, its lack of strategic planning and limited representation power restrict its applicability to complex protein design tasks. Reinforcement learning methods augmented with advanced protein language models, as demonstrated in our framework, provide a more scalable and effective solution for navigating the protein sequence space.

Limitation. Our approach faces two primary constraints. First, the computational burden of generating ESM2 embeddings for every MCTS state evaluation creates significant overhead. ESM2 inference is orders of magnitude slower than processing one-hot encodings, limiting real-time exploration—especially for long sequences (>500 residues) where GPU memory constraints further restrict batch sizes. While embeddings enrich representations, this trade-off may hinder large-scale searches. Second, architectural simplifications risk underutilizing ESM2’s capabilities. Though ESM2

embeddings provide global context, the lack of tailored inductive biases (e.g., position-sensitive decoders) may weaken mutation effect modeling. Additionally, the surrogate model’s dependence on wet-lab data—while mitigated by ESM2’s transfer learning—still necessitates expensive experimental validation for reward calibration.

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